



Associations Between Neurophysiological and Cognitive Performance in Patients with Psychosis: an Analysis of the N1, P2 and MMN Event Related Potentials

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Background

- Both cognitive performance and neurophysiological Event Related Potential (ERP) components have been shown to be altered in people with psychosis
- Prior studies have shown that greater symptom severity is associated with more impaired P3 ERP¹; greater P3 amplitude is associated with better verbal memory performance²; and that MMN impairment is not associated with symptom severity³. However, less is known about the relationships between clinical symptoms and N1 or P2 ERP components.
- We examined relationships between specific domains of cognition (MATRICS), symptom severity (PANSS), and three ERPs (N1, P2 and MMN) in 103 patients with psychosis to determine whether specific ERP phenotypes correlate with cognitive domains, and/or symptom severity.

Demographics

Table 1
Demographic Information

	SZ spectrum N = 30	BPI N = 73	p-value
Sample size			
Sex (%)			.048*
Female	33.33	54.79	
Male	66.67	45.21	
Mean age ± SD (years)	40.53 ± 11.01	29.63 ± 8.15	.000**
Race (%)			.765
Caucasian	83.33	80.82	
Other	16.67	19.18	
Mean education ± SD (years)	14.60 ± 1.98	15.12 ± 2.05	.238

* $p < .05$; ** $p < .001$

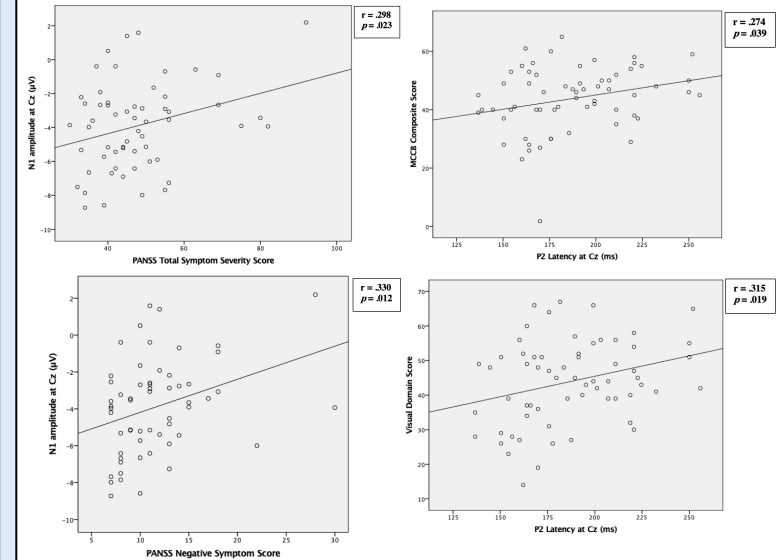
Methods

- Cognition** was assessed with the MATRICS Consensus Cognitive Battery (MCCB) including speed of processing, attention/vigilance, working memory (nonverbal & verbal), verbal learning, visual learning, reasoning and problem solving, and social cognition domains and composite scores
- Symptom severity** was assessed with the Positive And Negative Syndrome Scale (PANSS) including subdomains for positive, negative, and general symptom severity
- ERP data** (N1, P2 & MMN) were collected via EEG recording (see ⁴)
 - N1 and P2 ERPs, elicited using an auditory oddball paradigm, reflect perceptual information registration and processing
 - MMN, elicited using a duration MMN paradigm, reflects an automatic cortical change-detection process to discriminate changes in a stream of auditory stimulation
- Statistical analyses** included Pearson's correlations, linear regression and independent samples *t*-tests. Symptom severity variables was included as covariates.

Results

- The visual domain and composite score were significantly correlated with the **P2 latency** (Figure 1)
 - Increased **P2 latency** was associated with better visual learning ($r = .315$; $p = .019$) and overall cognitive composite score ($r = .274$; $p = .039$).
- Negative and overall symptom severity correlated with the **N1 amplitude**
 - Reduced **N1 amplitude** was associated with greater negative symptoms ($r = .330$, $p = .012$) and overall greater symptom severity ($r = .298$, $p = .023$).

Figure 1



Discussion

- Our results support the **N1/P2 ERP being involved in attention-modulated processes** required for cognitive function
- Longer P2 latency may index **greater attentional resources required** for stimulus classification processing
- Reduced N1 amplitude is closely associated the expression of psychotic symptoms
- There were no effects of diagnosis on any EEG or cognitive measure